Technical Self-Assessment of My UAV-IoT Coverage Research

What I implemented, what my experiments support, and what I need to fix

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Executive summary

My project's strongest contribution is the **feasibility-certified evaluation framework**, not the MAPPO controller itself. My current experiments show that a hand-designed reactive rule substantially outperforms my MAPPO implementation under the tested setup. I should therefore frame this work as a methodological evaluation paper with an honest negative multi-agent reinforcement learning (MARL) result, rather than as a claim that I have developed a superior learned UAV controller.

The main value is that I separate **task infeasibility from controller failure**. A packing-based lower bound exposes a configuration that passes a simple area-surplus check but cannot achieve full regional K-coverage. Combined with a coverage-achievability oracle, non-degenerate metrics, and reproducibility audits, this provides a stronger basis for interpreting experiments. At the same time, I must limit conclusions about MAPPO because the current training budget is small, clarify where a centralized safety filter is used, and make the baseline information access explicit.

My central claim: I built an evaluation and auditing framework that reveals when UAV coverage experiments are infeasible or misleading, and I used it to show that the tested MAPPO controller underperforms a reactive rule on a corrected feasible persistent K-coverage task. This is a meaningful research contribution, but it does not prove that MAPPO cannot solve the problem or that the diagnosed cause is definitively credit assignment.

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1. My high-level verdict

What I did well

My report has several unusually strong aspects for an R&D/student project:

1. **I discovered that the original configuration was infeasible.**
   The observation that the conventional area-surplus criterion passes while the corner-packing bound rejects the configuration is important. This is one of the best parts of the work.
2. **I separated "controller failure" from "task infeasibility."**
   Many UAV/RL papers report coverage ratios without proving that full coverage was ever achievable. My oracle/certificate idea directly addresses that weakness.
3. **I reported a negative result honestly.**
   I did not hide that the reactive rule beats MAPPO. That is scientifically stronger than forcing a weak learning result to look good.
4. **I added serious audit/reproducibility machinery.**
   Automated claim audits, deterministic seed handling, path-loss checks, and negative controls are far above typical course-project quality.
5. **I correctly identify credit assignment as a likely bottleneck.**
   The result that a shared team reward may fail in persistent multi-UAV coverage is plausible and connects to known MARL credit-assignment issues. COMA, for example, was explicitly motivated by multi-agent credit assignment and uses a counterfactual baseline to isolate an agent's contribution [1].

What remains weak or risky in my work

1. **The novelty is overclaimed in places.**
   The packing bound itself is not a new mathematical invention; it is a standard covering/packing lower-bound idea. My innovation is its **application as an evaluation certificate for UAV-IoT persistent K-coverage**, not the lemma itself.
2. **The MAPPO result is too undertrained to support broad claims.**
   25 episodes / 8 seeds is not enough to conclude "MAPPO is bad for this problem." I can conclude: **under this implementation, training budget, reward, observation, and architecture, MAPPO underperformed a reactive rule.**
3. **The "decentralized execution" claim is weakened by the global safety/action mask.**
   If the connectivity mask uses global true state at execution time, then the actor is decentralized but the system is not fully decentralized. I should call it:
   "decentralized actor with centralized safety-filter / supervisor."
4. **The reactive heuristic may not be a fair opponent unless carefully described.**
   If it uses exact coverage deficits, exact sensor states, or global geometry, then it may be more privileged than the learned policy. My "local assignment" version helps, but reviewers will still ask whether the heuristic gets cleaner information than MAPPO.
5. **The physical model is over-detailed relative to how much it affects the result.**
   My own audit says the link margin is huge, so the PHY is not the binding constraint. That is fine, but I should avoid implying that PHY realism is central to the result.

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2. Is my work innovative or already established?

My short answer

Partly innovation, partly recombination, partly rediscovery.

The pieces individually already exist:

UAV coverage with energy-aware DRL already exists.

Persistent UAV coverage with recharging already exists.

Action masking in RL already exists and is formally studied; invalid action masking is known to be a valid technique in policy-gradient algorithms [2].

MAPPO is a known cooperative MARL baseline and has been shown to work well on standard multi-agent benchmarks [3].

Joint UAV coverage, connectivity, and charging existed before my work; Trotta et al. propose centralized and distributed strategies for stationary UAV coverage, connectivity, and energy replenishment [4].

K-coverage and recharging in UAV-IoT networks is also studied; Song et al. explicitly address assigning UAVs for K-coverage and recharging in IoT networks [5].

Persistent multi-UAV coverage with priority, outage memory, finite battery, recharging, and MARL is close to my topic; a recent POGS-QMIX paper studies persistent geometric multi-UAV service coverage with finite battery and recharging, using priority/outage guidance and return-energy masks [6].

Where my work is more original

My strongest originality is **not** "I use MAPPO for UAVs." That is not enough.

My strongest originality is:

A feasibility-certified evaluation protocol for persistent redundant UAV coverage, combining a packing-number lower bound, a teleporting oracle ceiling, and non-degenerate metrics to avoid misreading infeasible experiments.

That is a legitimate research angle.

I should phrase the contribution carefully:

I should **not** say: “I invented packing-based feasibility.”

I should instead say: “I integrate a packing-based infeasibility certificate into the evaluation protocol for persistent K-coverage UAV-IoT experiments, and show that a commonly used area-surplus criterion can be misleading.”

That framing is much more defensible.

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3. Strongest parts of my report

3.1 Feasibility certificate is genuinely valuable

My argument:

$$P(A,R)=max\{|X|:X\subseteq A,\ \|x-x'\|>2R\}$$

Then any K-cover requires:

$$N\geq K\,P(A,R)$$

This is correct as a lower bound. The logic is simple and strong: if two points are more than $2R$ apart, one coverage disk cannot cover both; therefore each packed point needs K distinct disks.

For my square region:

Region: 80×80

$R_{cov}=35$

$2R_{cov}=70$

Four corners are pairwise farther than 70 along sides/diagonals.

Therefore $P(\Omega,35)=4$.

For $K=2$, $N\geq 8$.

This directly exposes the problem with the old $N=5, K=2, R=35$ configuration. That is a meaningful finding.

3.2 I correctly criticize area-surplus

The area-surplus criterion:

$$\sigma=\frac{N\pi R_{cov}^2}{K|\Omega|}$$

can be necessary-ish for region coverage but is not sufficient. My example is good:

$N=5, K=2, R=35$

$\sigma\approx 1.503$

Yet the region cannot be K-covered because the corners require 8 UAVs.

This is an excellent teaching example, and I should keep it.

3.3 My critique of the metrics is sound

The “service continuity” metric becomes uninformative when no feasible configuration exists. If the feasible set is empty, every policy gets zero, so the metric cannot distinguish:

terrible controller,

almost-correct controller,

impossible task.

My use of graded metrics like persistent coverage, under-coverage rate, and refresh time is correct.

I discuss the logical caveat in Section 5 below.

3.4 My honest negative result is valuable

My main experimental result is:

Reactive rule: around $C_K \approx 0.953$ to 0.972

MAPPO: $C_K \approx 0.674$

Gap: about 0.298, 95% interval $[0.248, 0.349)$

This is not a failure of the project. It is actually one of the project's most research-like outcomes, because it says:

"A standard CTDE MAPPO setup did not solve this persistent redundant UAV coverage problem as well as a simple reactive rule."

That is useful when I frame it correctly.

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4. Main technical weaknesses I need to address

4.1 I should not present MAPPO as the successful proposed solution

My current paper framing sometimes sounds like:

"I propose a MAPPO controller for persistent coverage."

But my results say:

"MAPPO loses badly to the reactive rule."

So I should change the paper's framing.

My framing should be:

"I propose a feasibility-certified evaluation framework and use it to audit a CTDE MAPPO controller. The audit reveals that, on the corrected feasible configuration, a hand-designed reactive rule substantially outperforms MAPPO."

This framing is much stronger and more honest.

4.2 My training budget is too small for a strong MARL conclusion

A reviewer will attack this immediately.

I report roughly 8 training seeds and 25 episodes. For MAPPO, that is very small. MAPPO papers usually train over large numbers of environment steps and perform careful hyperparameter tuning. The original MAPPO work reports strong results across standard cooperative benchmarks, but that strength depends on careful implementation and sufficient training [3].

So my conclusion should be limited:

Bad claim:

"MAPPO cannot solve persistent UAV-IoT coverage."

A better claim is:

"Under the tested observation, reward, masking, and training budget, my MAPPO implementation does not match a reactive rule."

An even better claim is:

"This motivates future work on per-agent credit assignment, stronger architectures, curriculum learning, and centralized/MPC baselines."

4.3 My reactive rule may be unfairly strong

The reactive rule seems to know where under-covered sensors are and moves toward them. That is exactly the high-level structure the learned policy struggles to discover.

If the heuristic uses global sensor coverage state, while MAPPO receives a compressed deficit grid, then the comparison is not purely "learning vs rule." It is partly:

exact symbolic state access,

handcrafted assignment,

privileged geometry,

deterministic reactive behavior,

versus a learned decentralized policy with partial observation.

My local-assignment reactive variant helps a lot, especially because it still performs extremely well. But I need to be very explicit:

What information does my reactive rule receive?

Does it receive the same information as MAPPO?

Does it use the true per-sensor $n_d(t)$?

Does my reactive rule know all sensor positions?

Does my reactive rule know global UAV positions?

Does my reactive rule use centralized assignment?

If so, I should call it a **strong oracle-like heuristic baseline**, not a fair decentralized peer.

4.4 "Decentralized execution" is overstated

My actor may be decentralized, but the safety filter/mask appears to use the simulator's true global state to reject actions that would disconnect the swarm.

That means execution is not purely decentralized.

My terminology should be:

"CTDE actor with centralized safety-filtered execution."

or

"Decentralized policy executed under a centralized feasibility shield."

This matters because a reviewer could argue that if the mask needs global connectivity, then my deployed system still requires a central supervisor.

4.5 My connectivity model ignores sink connectivity

I say that if the UAV graph is connected, sensor data can reach the sink, but later admit that the sink is not a vertex and is assumed to remain within communication range of at least one UAV.

That is a significant modeling assumption.

For my UAV-IoT communication network, the graph should include:

UAV nodes,

sink/base-station node,

maybe sensor-to-UAV access links.

I should require the active UAV graph plus the sink to be connected. Otherwise, my model could count a connected UAV swarm as connected even when it is disconnected from the base station.

Proposed fix:

$$G_t = \left(V_{UAV} \cup \{sink\}, E_{UAV-UAV} \cup E_{UAV-sink}\right)$$

I should then evaluate sink reachability explicitly.

4.6 My physical-layer model is mostly non-binding

I include Al-Hourani LoS probability, path loss, 802.11 MCS gating, OMNeT++/INET audit, etc. The Al-Hourani model is a known air-to-ground model for optimizing LAP/UAV altitude for coverage [7]. 3GPP TR 36.777 is also a standard reference for enhanced LTE support for aerial vehicles [8].

My own audit shows huge link margins, so the radio model does not materially affect the central result. That is not bad, but I should state this clearly:

"At the chosen parameters, the radio gate is present but non-binding; geometry and fleet feasibility dominate."

Otherwise, the discussion risks making physical-layer realism sound more central than the evidence supports.

4.7 My energy and charging assumptions are not realistic enough

I correctly admit that the charging rate is about 3.6C and is a modeling assumption. That is good.

I should avoid claiming full realism and instead state:

"The energy model is a stress-test abstraction designed to make charging contention occur within a tractable episode, not a calibrated battery model."

Also, if the episode is 400s and hover endurance is 600s, the battery stress depends heavily on movement duty cycle and charge rate. A reviewer may ask: am I studying persistent operation, or just a short stress episode?

I can defend it, but I need to make the scope modest.

4.8 I need stronger baselines

The baselines are decent for a project, but not enough for a paper-level claim.

I should add or discuss:

1. **Centralized MPC / greedy assignment baseline**
   A centralized controller that directly optimizes current deficit, connectivity, and energy.
2. **MILP or CP-SAT placement/repositioning oracle for small cases**
   This would provide a stronger upper bound than a teleporting placement oracle.
3. **QMIX / VDN / QTRAN-style baselines**
   Persistent multi-UAV coverage with energy and outage guidance has been studied with QMIX-style value factorization; the POGS-QMIX work is especially close [6].
4. **COMA / counterfactual advantage baseline**
   Since I diagnose credit assignment, I should test at least one method designed for credit assignment. COMA explicitly addresses multi-agent credit assignment using a counterfactual baseline [1].
5. **Graph neural network / attention policy**
   UAV swarms have permutation and spatial structure. A shared MLP with one-hot identity may be weaker than a graph/equivariant policy. Recent UAV swarm coverage work explicitly uses symmetry-informed MARL for communication coverage [10].

4.9 My claim audit sounds stronger than it is

I say 38 items, 0 failing. But my abstract also says the audit does not audit Section VIII results.

This can confuse readers.

I should state the scope precisely:

“The audit verifies model, protocol, physical-layer algebra, feasibility, and determinism claims. It does not independently validate the statistical performance conclusions of Section VIII beyond recomputing artifact-generated numbers.”

Otherwise, the audit may sound as though it proves the experimental claims are scientifically correct, when it mainly shows that the reported numbers can be reproduced.

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5. Mathematical and logical issues I need to fix

5.1 I need to state the service-continuity logic precisely

I write something like:

$$SC(\pi)=0 \text{ for every policy } \pi \Leftrightarrow F=\varnothing$$

The spirit is right, but the logic needs care.

If $F\neq\varnothing$, a bad policy may still never visit F, so $SC(\pi)=0$ for that policy. I partly clarify this, but the equation can still be misread.

My corrected formulation should be:

$$F=\varnothing \Rightarrow SC(\pi)=0 \;\forall\, \pi$$

and

$$\max_{\pi} SC(\pi)>0 \Rightarrow F\neq\varnothing$$

But the reverse depends on reachability, motion constraints, energy, initial state, and horizon. A feasible static configuration does not automatically imply a feasible trajectory from the initial cluster under battery/connectivity constraints.

I should write:

“If the feasible set is empty, the binary service-continuity metric is identically zero for all policies. Conversely, observing zero service continuity does not prove infeasibility, because a poor policy on a feasible instance may also score zero.”

This wording is logically safer.

5.2 My teleporting oracle is an upper bound for coverage, not full service

I already mention this, but I should emphasize it.

A teleporting static placement oracle upper-bounds instantaneous coverage. It does not necessarily upper-bound:

recovery time,

energy-feasible persistence,

connectivity-maintaining trajectories,

charging contention,

service continuity over a trajectory.

I should call it:

“coverage achievability ceiling”

rather than saying:

“overall task optimum.”

5.3 I must separate region coverage from sensor-set coverage

This is one of the most important distinctions I need to preserve.

My original $N=5$ configuration is **provably infeasible for covering the entire region**. But my actual experiments evaluate **20 discrete sensors**. For a discrete random sensor set, $N=5$ may often be feasible.

I do acknowledge this, which is good. I should make this distinction explicit throughout:

Region certificate: $N=5$ impossible.

Sensor-set certificate: depends on actual layout.

My random-draw result: $N=5$ excluded for 13% of 30 layouts.

I must not conflate these.

5.4 I need to qualify the tightness construction using all constraints

I say $N=8$ is exactly sufficient for region K-coverage. Geometrically, yes.

But my full system also has:

safety distance,

communication range,

radio gate,

no-articulation / H-hop mask,

starting cluster,

finite battery,

charging,

reserve behavior.

I should distinguish between:

“Geometrically tight for static region K-coverage.”

from:

“Dynamically achievable under the simulator.”

I state that my audit checks connected placement and safety-mask admissibility. I should make this explicit immediately after the tightness claim.

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6. How I position my work in the literature

What is already known and is not novel

MAPPO itself is not my contribution

MAPPO is not my contribution. It is a known algorithm. The standard reference shows PPO-style methods can be strong in cooperative multi-agent benchmarks [3].

Action masking is not novel

Action masking is not novel. It has been formally studied in policy-gradient algorithms and shown to produce valid gradients under the masking setup [2].

Time-limit truncation handling is not novel

My distinction between termination and truncation is correct, but not novel. Pardo et al. discuss why bootstrapping should be handled carefully when episodes end because of time limits [9].

UAV coverage and charging are already studied

Trotta et al. already jointly consider coverage, connectivity, and charging strategies for distributed UAV networks, including centralized optimal and distributed scheduling/positioning approaches [4].

K-coverage and recharging are already studied

Song et al. study methods to assign UAVs for K-coverage and recharging in IoT networks [5].

Persistent energy-aware multi-UAV coverage with MARL is closely related

Recent work such as POGS-QMIX studies persistent geometric multi-UAV service coverage with finite batteries, recharging, outage memory, return-energy action masks, multi-seed evaluation, and sensitivity studies [6].

What may be novel in my work

My novelty is the **combination of evaluation devices**:

packing-based infeasibility certificate,

corrected feasible configuration,

teleporting coverage oracle,

non-degenerate metrics,

explicit demonstration that area-surplus and binary service-continuity can mislead,

audit-backed negative result for MAPPO.

This is a defensible research niche.

I should not claim that I am the first to combine coverage, charging, connectivity, failure, and MARL unless my literature search is very strong. I should phrase the novelty claim as:

“I am not aware of prior work combining these five elements with a feasibility-certified evaluation protocol.”

That is safer.

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7. Paper-writing issues I need to address

7.1 My report is too defensive

The writing repeatedly says things like:

“I do not claim…”

“the framework earns its keep…”

“the difference between a number and a measurement”

“the cheapest test does not resolve it”

“this is not cosmetic”

This is intellectually honest, but the tone can become argumentative. I should keep the academic writing calmer.

I should replace this with:

“The framework is the difference between a number and a measurement.”

with:

“The framework provides the feasibility context needed to interpret reported coverage values.”

I should replace this with:

“The area-surplus criterion certifies nothing.”

with:

“The area-surplus criterion is not sufficient for pointwise K-coverage.”

7.2 I list too many contributions

I list 11 contributions. That is too many.

I should consolidate the contribution list into 4–5 items:

Unified UAV-IoT persistent K-coverage model with energy, connectivity, charging, and failure recovery.

Feasibility-certified evaluation framework using packing lower bound and teleporting coverage oracle.

CTDE MAPPO controller with safety mask, charging arbitration, and reserve recovery.

Non-degenerate evaluation protocol and reproducible audit.

Negative empirical result: reactive rule outperforms MAPPO; ablations diagnose possible causes.

7.3 How I should describe empirical findings

I should use terms such as:

“measured,”

“observed,”

“under the tested setting,”

“consistent with,”

“does not rule out.”

I already often do this, but I should apply it consistently.

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8. My experimental limitations and concerns

8.1 Eight seeds is okay for a project, weak for strong claims

The main gap is large enough that 8 seeds may be persuasive. However, I should treat ablations with confidence intervals crossing zero as inconclusive.

For example:

identity: +0.008, CI crosses zero,

local critic: -0.035, CI crosses zero,

threshold term: interval crosses zero.

My conclusion should be:

“Not resolved at this budget.”

rather than saying:

“does not matter.”

8.2 The budget sweep does not exclude training budget

I say training from 8 to 25 episodes improves C_K by +0.086, which accounts for only 29% of the gap.

That does **not** prove that a larger budget cannot close the gap because learning curves can be nonlinear. The supported conclusion is:

“Within the tested budget range, increased training did not close the gap.”

8.3 I need to establish hyperparameter-tuning fairness

Did I tune MAPPO as heavily as I tuned the reactive rule? If not, reviewers may consider the comparison unfair.

At minimum, I should report:

actor/critic network sizes,

learning rate,

entropy coefficient,

rollout length,

minibatches,

PPO epochs,

reward normalization,

value normalization,

gradient clipping,

action mask implementation,

number of environment steps,

number of policy updates.

I likely already have much of this, but I should make it easy to find.

8.4 I need to report wall-clock and sample-compute cost

If reactive rule has no training cost and beats MAPPO, that is a useful engineering result. I should also report:

training time,

inference time,

simulation speed,

number of environment transitions.

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9. What I should change before submitting

What I must fix

I should reframe the paper around evaluation methodology, not MAPPO success.

I should soften my novelty claims.

I should clarify decentralized execution versus the centralized safety filter.

I should separate region feasibility from sensor-set feasibility throughout the paper.

I should fix the service-continuity logic.

I should state that the teleporting oracle is a coverage ceiling, not the full task optimum.

I should reduce the contribution list to 4–5 items.

I should define exactly what information each baseline receives.

I should state stronger limitations around the training budget and seed count.

I should model sink connectivity or clearly mark its omission as a simplifying assumption.

What I should add if possible

One credit-assignment baseline: COMA, difference reward, HAPPO/HATRPO, or counterfactual advantage.

One graph/equivariant policy baseline.

One centralized MPC/assignment heuristic.

A longer MAPPO training run, even if only for 2–3 seeds, to show whether learning curves plateau.

A table comparing information access for each method.

I can make the baseline information-access comparison explicit as follows:

MAPPO: Global sensor positions are partial/broadcast; the under-covered set is represented through a compressed deficit grid; global UAV graph information is available only through the mask; it learns; its actor is decentralized; and centralized safety filtering is used.

Reactive-global: It uses global sensor positions, the exact under-covered set, and the global UAV graph; it does not learn; it is a global/centralized heuristic; and centralized safety is used.

Reactive-local: It uses local sensor positions/deficits and local state; access to the global graph is uncertain; it does not learn; it may be a local rule; and centralized safety is used.

I should include this comparison to make the fairness issue transparent.

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10. My final assessment

Is my project valid R&D?

Yes. My project is valid R&D.

Is my project publishable as currently framed?

Probably not yet, because my claims are too broad and my MAPPO result is weak.

Could my project become a good paper?

Yes, if I frame it as:

"Feasibility-certified evaluation for persistent redundant UAV-IoT coverage: why naive coverage metrics mislead, and what a MAPPO negative result teaches me."

That is much stronger than:

"A MAPPO framework for fault-tolerant UAV-IoT coverage."

What is my actual contribution?

My actual contribution is:

I built a rigorous evaluation and auditing framework that reveals when UAV coverage experiments are infeasible or misleading, and I used it to show that a standard MAPPO controller underperforms a reactive rule on a corrected feasible persistent K-coverage task.

That represents real research value.

What have I not yet proven?

I have **not** proven:

that MAPPO cannot solve the task,

that my learned controller is competitive,

that credit assignment is definitely the cause,

that my physical model materially changes the conclusion,

that my approach is fully decentralized,

that no prior work has combined similar elements.

My one-sentence summary

This is a strong evaluation-methodology project with an honest negative MARL result; its main innovation is the feasibility-certified measurement framework, not the MAPPO controller.

Reference links for this assessment

COMA: Counterfactual Multi-Agent Policy Gradients. URL: https://arxiv.org/html/1705.08926v3

Invalid action masking in policy-gradient algorithms. URL: https://arxiv.org/abs/2006.14171

The MAPPO cooperative multi-agent benchmark study. URL: https://proceedings.neurips.cc/paper_files/paper/2022/hash/9c1535a02f0ce079433344e14d910597-Abstract-Datasets_and_Benchmarks.html

Trotta et al.: UAV coverage, connectivity, and energy replenishment. URL: https://genesys-lab.org/papers/UAV-TransRobotics18.pdf

Song et al.: UAV assignment for K-coverage and recharging in IoT networks. URL: https://ieeexplore.ieee.org/document/10077427/

POGS-QMIX: persistent multi-UAV service coverage with finite batteries and recharging. URL: https://www.mdpi.com/2504-446X/10/9/688

Al-Hourani et al.: air-to-ground path-loss and coverage model. URL: https://ieeexplore.ieee.org/iel7/5962382/6975315/06863654.pdf

3GPP TR 36.777: enhanced LTE support for aerial vehicles. URL: https://www.3gpp.org/dynareport/36777.htm

Pardo et al.: time-limit truncation and bootstrapping. URL: https://arxiv.org/abs/1712.00378

Symmetry-informed MARL for UAV swarm communication coverage. URL: https://ieeexplore.ieee.org/document/10935710/